History Packaging Policies for LLM Recommendation: Evidence from Real LLM Scoring Under Prompt Budgets*

Wooseung Kang and Sang-Min Choi†

Gyeongsang National University, South Korea
{kung0615, jerassi}@gnu.ac.kr

Abstract

Large-language-model recommenders must decide which parts of a user's history survive under a fixed prompt budget. That packaging choice changes both ranking quality and the input-token burden, a deployment-facing proxy for latency, throughput, and operating expense on e-business ranking surfaces [5, 2, 12]. This paper studies the problem as a factorial design under real LLM scoring: Selection policy $\{\text{CC}, \text{Recency}, \text{Random}\} \times$ retained history length $K \in \{4, 8, 16, 32\}$. The main evidence layer is a 12-cell matrix on MovieLens-1M and BookCrossing, evaluated over seeds 42/43/44 with 512 users per dataset. One scope is one dataset-seed pairing, so the paper analyzes 6 scopes and 72 scope-level observations in total. The strongest evidence-backed findings are fourfold. First, selection is a first-order factor in the descriptive ANOVA-style decomposition: the eta-squared-style contributions are 0.1337 for selection, 0.0487 for retained length, and 0.0225 for the interaction. Second, the dataset split matters. On BookCrossing, CC exceeds Recency at every matched K (+0.0131 to +0.0238 mean AUC), with the best observed cell at CC-K-16 (0.676261 AUC, 348.20 tokens). On MovieLens-1M, Recency exceeds CC for $K \in \{4, 8, 16\}$ and becomes effectively tied at $K = 32$, with the best observed cell at Recency-K-16 (0.777892 AUC, 297.35 tokens). Third, paired tests show Random is often weaker than CC or Recency, but CC-versus-Recency is condition-dependent rather than one-sided. Fourth, saturation is policy-specific: CC and Recency flatten or reverse at high K, whereas Random improves monotonically with K in this bundle while remaining below the strongest CC/Recency cells. We therefore avoid single-winner narratives and frame the contribution as a controlled empirical study of selection-policy and retained-length interactions under a shared scoring protocol, not as a new recommender architecture or a production-latency benchmark. For e-business recommendation, the practical contribution is to treat history packaging as a deployment knob: low-token Recency cells expand feasible operating regions when serving budgets are tight, while CC-K-16 is attractive when higher-quality ranking justifies additional prompt cost.

1 Introduction

Large language models are becoming an increasingly common substrate for recommender systems because they can consume rich metadata, support flexible prompting, and unify retrieval, ranking, and explanation inside a single framework [15, 9, 17]. At the same time, recommendation remains a long-history decision problem. A practical system must choose which parts of a user's history should survive into the final scoring context.

This is where prompt budget becomes a systems and e-business issue rather than a cosmetic prompting detail. User histories are often much longer than the context a practical model can score repeatedly. More context can help, until it does not. Tokens cost money, longer prompts slow inference, and naive truncation can remove exactly the behavioral evidence that personalization needs. On high-volume

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†Corresponding author.

ranking surfaces, those prompt decisions propagate into service cost, throughput ceilings, and latency headroom, which are core concerns in value-aware recommender deployment [5, 2, 12]. The central design problem is therefore to determine which evidence should remain in context, how it should be structured, and which retention rule should govern selection under a fixed prompt budget.

Existing LLM-based recommenders attack this challenge through several routes. Retrieval-oriented systems such as ReLLa compress long histories into candidate-conditioned subsets before prompting [7]. Other approaches combine instruction tuning, reranking, collaborative filtering augmentation, or profile construction to improve alignment with recommendation tasks [15, 9, 13, 16, 14]. Long-context and memory-oriented studies make a related point from another angle: simply exposing more context does not guarantee better behavior because selection, position, and packaging also matter [8, 10, 6, 1, 4].

However, one important part of this design space is still underexplored. Once the backbone and the candidate protocol are held fixed inside a single recommendation framework, the effect of the history-packaging rule itself remains unclear. Existing work often changes both the framework and the packaging strategy at the same time, which makes it difficult to isolate whether the gain comes from the backbone, the retrieval recipe, or the packaging policy.

This paper studies that problem inside one synchronized recommendation framework. We fix one shared candidate protocol and one frozen LLM scorer, then evaluate a factorial design over packaging policies: Selection $\in \{\text{CC}, \text{Recency}, \text{Random}\}$ and retained length $K \in \{4, 8, 16, 32\}$. The main evidence is a 12-cell matrix on MovieLens-1M and BookCrossing, evaluated across seeds 42, 43, and 44 with 512 users per dataset. One scope is one dataset-seed pairing, so the paper analyzes 6 scopes and 72 scope-level observations; this is distinct from the underlying $512 \times 2 \times 3 \times 12$ user-cell evaluations used to compute those scope means. We do not claim a new state-of-the-art recommender, a new semantic retriever, or broad cross-family superiority over external systems such as ReLLa or RALLRec+. Instead, the methodological contribution is a controlled measurement design for a deployment question: under real LLM scoring, how do selection policy and retained length interact in the quality–cost tradeoff?

The answer is already more nuanced than a pooled single-winner story. In BookCrossing, CC is consistently above Recency at matched K, and CC-K-16 is the strongest observed cell. In MovieLens-1M, Recency is stronger for $K \in \{4, 8, 16\}$ and effectively tied with CC at $K = 32$, with Recency-K-16 as the strongest observed cell. A cautious reading is that, under the tested slices and shared scorer setup, BookCrossing benefits more from candidate-conditioned lexical or topical matching, whereas MovieLens-1M benefits more from short-horizon recency signal. That split is operationally important because it changes which frontier point is attractive under a given cost budget.

This empirical framing responds to a practical gap: many systems must choose a packaging rule before they can afford broader model or infrastructure sweeps. The contributions of this paper are as follows:

- We formulate prompt-budgeted history packaging in a synchronized recommendation framework as a factorial design over selection policy (CC, Recency, Random) and retained history length ($K = 4, 8, 16, 32$), clarifying the workflow from history retention to LLM scoring and cost accounting.
- We define a controlled evidence stack in which the main layer is a real-LLM-scored 12-cell matrix over two datasets, three seeds, and 512 users per dataset, and we make the scope accounting explicit: 6 dataset-seed scopes and 72 scope-level observations.
- We show that the pooled result hides a meaningful interaction: BookCrossing favors CC across matched K, while MovieLens-1M favors Recency through $K = 16$ and only converges at $K = 32$.
- We show that quality–cost frontiers and saturation behavior are policy-dependent: Recency-K-8 is a low-token frontier point, CC-K-16 reaches the highest pooled mean AUC, and later-K gains

flatten for CC/Recency while Random continues to improve monotonically from a weaker starting point.

2 Related Work

Recent work has shown that large language models can support recommendation, but long interaction histories remain difficult to use well. Retrieval-oriented methods such as ReLLa focus on extracting compact candidate-conditioned subsets of user behavior before final prompting [7]. Other systems study LLM-based recommendation more broadly, including recommendation-specific instruction tuning, retrieval-reranking pipelines, collaborative-filtering augmentation, and cold-start settings [15, 9, 13, 16, 14].

Our paper is narrower than those efforts. We do not study broad leaderboard performance or agentic recommendation behavior. Instead, we isolate a specific systems question that remains practically important when prompt budgets are fixed: how should a recommender package user history before scoring? We also keep the claim boundary narrow. ReLLa and RALLRec+ provide context for the broader space, but our empirical claim is internal to one synchronized recommendation setup with a shared candidate protocol and frozen scorer.

That question also connects to a broader observation from long-context LLM studies. Simply exposing more context does not guarantee better behavior, and representation, position, and packaging matter [8]. Recent profile and memory-modeling papers make a related point by treating long behavior sequences as a structured memory problem rather than as a simple prompting problem [10, 6, 1, 4]. Our saturation analysis is therefore theory-oriented but bounded: when later retained items add smaller gains than earlier ones, the pattern is consistent with diminishing returns from additional behavioral evidence rather than evidence that more history is universally better.

The paper also speaks to the e-business and value-aware recommendation literature. Economic recommender analyses emphasize that ranking quality must be read jointly with cost, service constraints, and business value rather than as an isolated offline metric [5, 2]. Work on profit-aware recommendation makes the same operational point from a different angle: ranking decisions are entangled with system objectives and resource tradeoffs [12]. In addition, work on choice overload shows that more options or more information do not automatically improve user outcomes [11]. We position our paper inside that operational perspective: retained history length is a scarce resource, and the empirical question is which packaging policy uses that resource most effectively under a fixed scoring contract.

3 Problem Formulation

We study prompt-budgeted recommendation for a user u with interaction history $H_u = (i_1, \ldots, i_n)$ and candidate item c. The key question is not whether more history exists, but how that history should be packaged before scoring when prompt cost is limited. The paper evaluates packaging policies under a shared candidate protocol and scorer, so differences can be attributed to the retention rule rather than to a different ranking backbone.

Let $\tau(i)$ denote the token set extracted from the title and category fields of item i. Following the implementation, the history query set is built from the most recent $\kappa = 3$ interactions:

$$Q(H_u) = \bigcup_{t=\max(1,n-\kappa+1)}^{n} \tau(i_t), \tag{1}$$

with fallback to the union over all history tokens if the recent set is empty.

For a candidate c, the synchronized framework ranks each history item by candidate-conditioned token overlap:

$$r(i,c \mid H_u) = |\tau(i) \cap (Q(H_u) \cup \tau(c))|. \tag{2}$$

History items are sorted by descending $r(i,c \mid H_u)$, with later interactions preferred on ties. Let $\pi(H_u,c)$ denote the resulting ranked order. The paper studies how different retention rules applied to $\pi(H_u,c)$ change quality and cost. In addition to the candidate-conditioned policy, we include recency-only and random-K packaging baselines so the results can distinguish informed packaging from naive truncation.

4 Factorial Packaging Policies and Evidence Layers

4.1 Packaging policies

All cells share the same candidate-conditioned ranking backbone; only selection policy and retained length vary. The candidate-conditioned rule is intentionally simple and lexical, not an embedding-based semantic retriever. This choice keeps the experimental contrast focused on packaging behavior under a transparent retention rule.

CC selection. CC keeps the top K items in the candidate-conditioned order:

$$S_{\mathrm{CC}}(H_u,c;K) = \mathrm{TopK}(\pi(H_u,c),K). \tag{3}$$

In the factorial bundle, $K \in \{4,8,16,32\}$.

Recency selection. Recency ignores candidate-conditioned ranking and keeps the most recent K interactions:

$$S_{\mathrm{Recency}}(H_u;K) = \{i_{n-K+1},\ldots,i_n\}. \tag{4}$$

Random selection. Random samples K interactions from H_u under the experiment seed:

$$S_{\mathrm{Random}}(H_u;K) = \mathrm{Sample}(H_u,K). \tag{5}$$

Together these define the 12 canonical cells

$$\{\mathrm{CC},\mathrm{Recency},\mathrm{Random}\} \times \{4,8,16,32\}. \tag{6}$$

Figure 1 summarizes how the synchronized protocol turns a user history and candidate into comparable quality–cost evidence. The same workflow is used for every factorial cell; only the retention block changes.

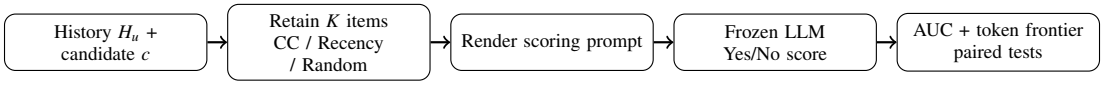


Figure 1: Workflow of the synchronized factorial protocol used in every cell; only the retention block changes.

Why adaptive variants are excluded. Adaptive operating-point variants are outside the paper's core factorial design. The reason is methodological rather than rhetorical: those variants change the retention rule itself across conditions, so they do not preserve a clean 3×4 comparison in which only selection family and retained length vary. Excluding them keeps the manuscript's main claims traceable to a balanced factorial surface with shared scorer, shared candidate protocol, and directly comparable cells.

4.2 Evidence layers and cost accounting

The main evidence layer uses one frozen Yes/No scorer, `Qwen/Qwen2.5-1.5B-Instruct`. It builds a prompt from the retained history and candidate and evaluates

$$s_{\text{llm}}(S,c) = \log P(\text{Yes} \mid \text{prompt}) - \log P(\text{No} \mid \text{prompt}). \tag{7}$$

We keep a leakage-free deterministic verifier as a supporting layer rather than a primary evidence layer because it offers a controlled approximation that is cheap to rerun and easier to inspect mechanistically. Selection still uses title and category tokens, but the deterministic verifier scores candidates from title tokens only. Let $\tau_{\text{ttl}}(i)$ denote whitespace-tokenized title tokens and $V_{\text{ttl}}(S) = \bigcup_{i \in S} \tau_{\text{ttl}}(i)$ the retained-history title vocabulary. The verifier is

$$s_{\text{det}}(S,c) = \sum_{w \in \tau_{\text{ttl}}(c)} \mathbb{1}[w \in V_{\text{ttl}}(S)]. \tag{8}$$

The factorial package reports the measured LLM input cost directly as `actual_input_tokens_avg`, i.e., the true scorer-side prompt token count after retention and candidate rendering. Additional overhead fields are retained only as auxiliary diagnostics; the headline quality–cost analysis uses measured input burden rather than proxy accounting.

5 Experimental Setup

5.1 Research Questions

We organize the paper around three research questions.

- **RQ1**: Under real LLM scoring, how strong are the selection-policy and retained-length effects in a factorial matrix?
- **RQ2**: Which factorial cells lie on the quality–actual-token Pareto frontier?
- **RQ3**: When do paired comparisons support CC over Recency, when do they support Recency over CC, and when is the evidence only mixed or tied?

5.2 Experimental Settings

5.2.1 Datasets

We use two CTR-style recommendation datasets. MovieLens-1M contains one million movie ratings from 6,040 users and 3,952 movies [3]. BookCrossing contains roughly 1.15 million book ratings from a large community of readers [18]. Both datasets are converted into CTR comparison slices by raw-interactions last-item holdout: all but the final interaction form the history and the final interaction becomes the positive target.

We prepare 512 users per dataset and evaluate seeds $\{42, 43, 44\}$. A user must have at least 20 interactions to enter the default slice. This yields two related counting levels that the paper keeps separate. At the user level, the scorer executes $512 \times 2 \times 3 \times 12$ user-cell evaluations. At the manuscript evidence level, one *scope* is one dataset-seed pairing, so there are $2 \times 3 = 6$ scopes. Each scope contributes 12 cell means, producing $6 \times 12 = 72$ scope-level observations for the descriptive factorial analyses.

5.2.2 Evaluation Protocol

Each CTR example contains one positive target and 19 negatives sampled without replacement from the global item pool after excluding the target and all history items; the 20 candidates are then shuffled. Our main analysis centers on AUC and measured actual input tokens because the paper asks a quality–cost question. Actual input tokens are a controlled cost proxy for the prompt burden; they should not be read as measured production latency, throughput, GPU memory, or online CTR. Paired significance tests use per-user AUC with Wilcoxon signed-rank tests and paired bootstrap intervals, and we use them to summarize directional stability rather than to make causal claims.

5.2.3 Implementation

The main matrix executes all 12 canonical cells (CC / Recency / Random $\times$ $K \in \{4, 8, 16, 32\}$) across both datasets and seeds $\{42, 43, 44\}$ using one frozen scorer, Qwen/Qwen2.5-1.5B-Instruct, throughout. Prompt cost is reported as measured actual input tokens, i.e., the true scorer-side token count after retention and candidate rendering, rather than any proxy accounting field.

6 Main Results

Table 1 and Figure 2 answer the main question under real LLM scoring for the 12-cell factorial design. The headline is not that one policy wins everywhere. The evidence supports a structured tradeoff surface whose shape depends on dataset, retained length, and cost. Pooled across scopes, CC-K-16 achieves the highest mean AUC (0.722262) at 329.77 measured input tokens, while Recency-K-8 reaches 0.713636 AUC at a substantially lower token cost (185.86). Random-family cells remain quality-limited across the matrix despite low-token variants.

The descriptive ANOVA-style decomposition confirms that policy choice is first-order in this bundle: relative to the grand mean AUC (0.696158), CC and Recency shift upward (+0.0160 and +0.0136), whereas Random shifts downward (-0.0296). The eta-squared-style shares show the same ordering, with selection (0.1337) larger than retained length (0.0487) and the interaction still material (0.0225). These shares are descriptive, not causal estimates. Their practical meaning comes from the induced operating decision: the interaction is smaller than the main policy effect, but it is large enough to reverse the preferred structured policy across datasets and to move the frontier point selected under a token budget.

The dataset split makes the CC-versus-Recency interaction concrete. In BookCrossing, CC exceeds Recency at every matched K: the mean CC-minus-Recency deltas are +0.018469 at $K = 4$, +0.013141 at $K = 8$, +0.023832 at $K = 16$, and +0.021399 at $K = 32$. Significance is limited at smaller K but becomes stronger at larger K, with CC significant in 2/3 seeds at $K = 16$ and 3/3 seeds at $K = 32$. In MovieLens-1M, the pattern flips through most of the range: the mean CC-minus-Recency deltas are -0.022752 at $K = 4$, -0.025271 at $K = 8$, -0.009629 at $K = 16$, and +0.000411 at $K = 32$, with Recency significant in all three seeds at $K = 4$, in 2/3 seeds at $K = 8$, in 1/3 seeds at $K = 16$, and no decisive winner at $K = 32$. Put differently, BookCrossing favors the candidate-conditioned rule across matched K, whereas MovieLens-1M favors recency until the highest tested retention level, where the two policies converge. A cautious observational interpretation is that, under the tested slices, BookCrossing gains

Table 1: Main real-LLM-scored factorial comparison over two datasets and seeds $\{42, 43, 44\}$ (12 canonical cells). Cells report mean AUC / measured actual input tokens. Dataset-separated columns make the interaction visible inside the manuscript.

Policy	K	BookCrossing	ML1M	Pooled
CC	4	0.663926 / 134.17	0.732165 / 122.66	0.698045 / 128.42
CC	8	0.668431 / 211.98	0.746710 / 186.57	0.707571 / 199.28
CC	16	**0.676261 / 348.20**	0.768263 / 311.34	**0.722262 / 329.77**
CC	32	0.670813 / 547.59	0.771090 / 540.47	0.720951 / 544.03
Recency	4	0.645456 / 128.36	0.754917 / 120.66	0.700187 / 124.51
Recency	8	0.655290 / 192.12	0.771981 / 179.59	**0.713636 / 185.86**
Recency	16	0.652429 / 317.70	**0.777892 / 297.35**	0.715161 / 307.53
Recency	32	0.649414 / 518.55	0.770679 / 519.22	0.710047 / 518.88
Random	4	0.591317 / 124.97	0.672406 / 120.40	0.631862 / 122.69
Random	8	0.611619 / 187.22	0.705284 / 179.52	0.658452 / 183.37
Random	16	0.629591 / 311.76	0.731873 / 297.73	0.680732 / 304.75
Random	32	0.643435 / 514.73	0.746539 / 520.12	0.694987 / 517.43

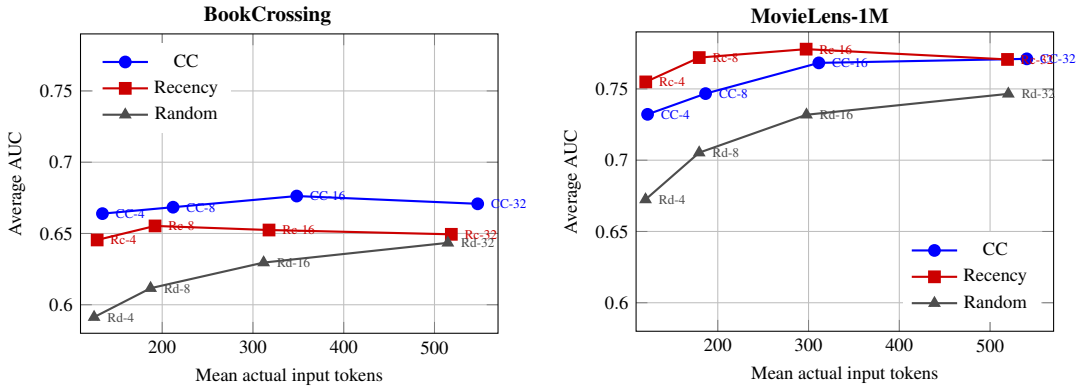


Figure 2: Dataset-separated quality–cost frontiers for the real-LLM-scored factorial matrix. BookCrossing shows a CC-led high-quality frontier, whereas MovieLens-1M shows a Recency-led frontier through $K = 16$; Random improves with K but remains below the strongest structured cells in both datasets.

more from candidate-conditioned lexical or topical matching, while MovieLens-1M gains more from short-horizon recency signal.

These dataset-separated cells also answer RQ2 more precisely. The pooled Pareto frontier contains Random-K-4, Recency-K-4, Recency-K-8, Recency-K-16, and CC-K-16, but the dataset winners differ. In BookCrossing, CC-K-16 is the best observed cell at 0.676261 AUC and 348.20 tokens. In MovieLens-1M, Recency-K-16 is best at 0.777892 AUC and 297.35 tokens, while CC-K-32 slightly exceeds CC-K-16 within the CC family but with much higher token cost. This is why a pooled frontier should be read as a deployment menu rather than as evidence for a universal policy ordering.

RQ3 asks whether paired tests support strong cross-family claims or only bounded ones. Across cross-family comparisons, CC-vs-Random is significant in 79/96 cases and Recency-vs-Random in 74/96, so Random is often weaker than either structured policy. By contrast, CC-vs-Recency is mixed overall (45/96 significant cases) and directionally depends on dataset, as the matched-K summaries above show. The paired evidence therefore reinforces the operating-region view instead of supporting a universal CC-or-Recency ordering.

The saturation evidence is also part of the main story. For CC, pooled marginal gains rise from +0.0095 AUC at $4 \rightarrow 8$ to +0.0147 at $8 \rightarrow 16$, then turn slightly negative at $16 \rightarrow 32$ despite a large token

Table 2: Saturation summary from the real-LLM-scored factorial matrix. Values report marginal AUC gains and marginal token cost by policy as K increases.

Policy	Marginal gains as K increases	Interpretation
CC	$4 \to 8$: +0.0095 AUC for +70.86 tokens; $8 \to 16$: +0.0147 for +130.49; $16 \to 32$: −0.0013 for +214.26	Best pooled quality appears at $K = 16$; the $K = 32$ step adds cost without added pooled AUC.
Recency	$4 \to 8$: +0.0134 AUC for +61.35 tokens; $8 \to 16$: +0.0015 for +121.67; $16 \to 32$: −0.0051 for +211.36	Earliest gain is efficient; later gains saturate fastest, which helps explain the low-token frontier points.
Random	$4 \to 8$: +0.0266 AUC for +60.69 tokens; $8 \to 16$: +0.0223 for +121.37; $16 \to 32$: +0.0143 for +212.68	Improves monotonically from a weak low-K baseline, but still remains below the strongest structured cells in absolute AUC.

increase. Recency behaves similarly but saturates earlier: +0.0134 at $4 \to 8$, only +0.0015 at $8 \to 16$, then -0.0051 at $16 \to 32$. Random, in contrast, improves monotonically (+0.0266, +0.0223, +0.0143) as K grows. We interpret that pattern cautiously. Random starts from a much weaker low-K baseline, so additional retained items can still recover omitted signal without making Random competitive with the strongest CC/Recency cells. The monotonic gains therefore do not overturn the main finding; they show that even a weak unstructured policy benefits from more retained evidence before its own returns begin to taper.

Taken together, the factorial view replaces single operating-point narratives with an operational region. Lower-token Recency-K cells are attractive when serving budgets are tight, while CC-K-16 is attractive when the deployment objective prioritizes peak pooled AUC and can absorb the extra prompt cost.

7 Saturation and Sensitivity Analyses

This section extends the main factorial results rather than replacing them. The paper's central evidence still comes from the real-LLM-scored matrix in Table 1, but saturation analysis helps explain why some frontier cells are efficient and why later-K expansions stop paying off.

7.1 Policy-specific saturation behavior

Table 2 summarizes K-step marginal gains from the factorial matrix. CC and Recency both show clear diminishing returns by $K = 32$, where extra token cost is large and the marginal AUC gain is near-zero or negative. This pattern is consistent with the idea that earlier retained interactions capture most of the useful behavioral evidence, while later additions become increasingly redundant or weakly aligned with the immediate scoring task. One compact way to formalize that intuition is through a diminishing-returns condition on retained-history information,

$$I(H_k; y) - I(H_{k-1}; y) \geq I(H_{k+1}; y) - I(H_k),$$

although we do not estimate this quantity directly in the present study. We present this as an observational interpretation rather than a proof.

The dataset split sharpens that explanation. In BookCrossing, CC improves through $K = 16$ and then declines at $K = 32$. Recency peaks earlier and turns negative sooner. In MovieLens-1M, both structured policies still gain through $K = 16$, but the later step is small for CC (+0.002827) and negative for Recency (-0.007213). These patterns align with the main-results interaction. The stronger policy is the one whose earlier retained items match the dataset's useful signal most efficiently.

Random requires separate interpretation. Its gains are monotonic in both datasets and in the pooled curve, but the increments still shrink with larger K. That is consistent with a weaker selector that begins by discarding too much useful evidence at low K; adding more random history partially repairs that information loss, yet the policy remains below the strongest CC/Recency cells in absolute AUC. The practical message is therefore not that Random is preferable, but that monotonic recovery can coexist with inferiority when the initial selection rule is weak.

These saturation trends explain the frontier geometry in Figure 2. Recency-K-8 is efficient because its earliest gain is high per token, while CC-K-16 is attractive when the objective emphasizes peak pooled AUC. The design implication is not to maximize K blindly, but to choose the operating region where additional retained history still earns meaningful quality gains per token under the target policy.

8 Discussion and Limitations

The evidence supports a narrow but trustworthy claim. The strongest stable result is not universal superiority for one packaging rule. It is that selection policy and retained history length jointly determine a structured quality–cost surface in the observed real-LLM matrix, and that the preferred structured policy changes across datasets. That is useful for system design because it supports frontier-oriented decisions instead of one-size-fits-all prescriptions.

This result also has a direct e-business interpretation. The measured input-token differences across frontier cells are not merely cosmetic prompt savings. Under serving stacks where inference burden scales approximately with prompt tokens, they translate into lower per-request operating cost, higher throughput under fixed infrastructure budgets, and a wider feasibility region for deploying LLM-based ranking on business-critical surfaces [5, 2, 12]. The paper therefore contributes a deployment-facing message: history packaging is an operational control variable, not only an offline modeling choice.

The saturation results matter in that operational framing because they constrain how far extra context should be pushed. More retained history is not automatically better, and the later-K flattening for CC and Recency is consistent with diminishing returns from additional user evidence. In that sense, the observed curves are also consistent with a diminishing-returns view of information selection, in which earlier retained items contribute larger marginal signal than later additions. That interpretation also resonates with adjacent work on overload and information efficiency in digital decision settings [11]. We intentionally keep the theoretical claim modest: the observed curves are aligned with diminishing-returns intuition, but the paper does not prove a general law for all recommenders or all prompt contracts.

The paper has several scope limits. First, the real LLM scorer is still one frozen evaluation contract, specifically `Qwen/Qwen2.5-1.5B-Instruct`, not the full space of possible recommenders or judge prompts. The primary scorer is a 1.5B-parameter instruction-tuned model, so whether larger open models, proprietary LLMs, or differently calibrated judge prompts exhibit the same packaging sensitivity remains an open question. Second, only two traditional recommendation domains are evaluated. This is sufficient for a controlled interaction study, but not for a generalization claim across news, retail, short-video, or session-based recommendation. Third, the CC policy is deliberately simple and lexical. More advanced semantic retrieval, embedding-based history selection, or hybrid profile construction could shift the frontier and should be compared in future work. Fourth, paired significance is strongest for comparisons against Random-family cells, while CC-versus-Recency comparisons are dataset-dependent and often non-significant, so cross-family claims must stay specific.

The cost and statistical readings should also remain bounded. Actual input tokens are measured from the scorer-side prompt, but they are still a proxy for deployment cost rather than direct measurements of latency, throughput, GPU memory, serving price, or online CTR. Likewise, the ANOVA-style output is descriptive rather than confirmatory causal inference, and the paired tests summarize directional stability under the tested user slices rather than a full mixed-effects or multiple-comparison-adjusted analysis.

These limits are deliberate: the goal is a clean, traceable contribution about history packaging under one synchronized recommendation contract rather than an overextended superiority claim. A natural next step is to replicate the same workflow across additional domains, stronger scorers, semantic retrievers, and production-oriented metrics.

9 Conclusion

This paper studies a systems question for recommendation under prompt constraints: how user history should be packaged before candidates are scored. Using a real-LLM-scored factorial matrix over MovieLens-1M, BookCrossing, seeds 42/43/44, and 512 users per dataset, we show that selection policy and retained length jointly shape a measurable quality–cost surface. The paper makes the evidence scale explicit: one scope is one dataset-seed pairing, giving 6 scopes and 72 scope-level observations across the 12 factorial cells.

The main conclusion is conditional rather than universal. CC-K-16 reaches the highest pooled mean AUC, but the dataset split shows that BookCrossing favors CC across matched K whereas MovieLens-1M favors Recency through $K = 16$ and only converges at $K = 32$. Random-family cells often improve as more history is retained, yet they usually remain weaker than the strongest structured cells. For e-business recommendation systems, this supports a practical workflow: treat history packaging as an operating-region decision, choose frontier cells according to token budget and quality target, and keep claims tied to the observed scorer contract instead of overgeneralizing beyond it. Future work should test the same workflow with broader datasets, larger or proprietary LLM scorers, semantic selection policies, and measured production metrics such as latency, throughput, serving cost, and online engagement.

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